



# MATLAB for Research

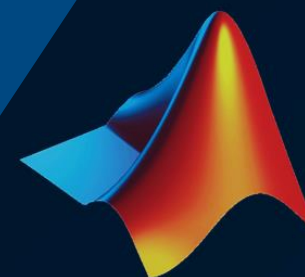
Computing, data analysis, visualization, & mapping



Laura Keen, PhD  
Senior Academic Engineer  
[lkeen@mathworks.com](mailto:lkeen@mathworks.com)



Paul Nihill  
Principal Account Manager  
[pnihill@mathworks.com](mailto:pnihill@mathworks.com)



MATLAB with

# Your MathWorks Team



**Paul Nihill**  
Principal Account Manager  
[pnihill@mathworks.com](mailto:pnihill@mathworks.com)  
(508) 647-2725



**Laura Keen, PhD**  
Senior Academic Engineer  
[lkeen@mathworks.com](mailto:lkeen@mathworks.com)



**Julie Casavant**  
Education Support  
Representative



**Bill Dickson**  
Customer Success  
Specialist

# Seminar Feedback

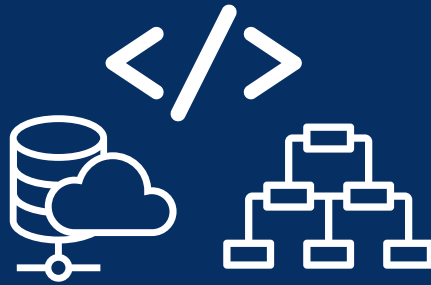
Enhancing Your MathWorks  
Seminar Experience



Enhancing Your MathWorks Seminar  
Experience – Fill out form



**Ease of Use**



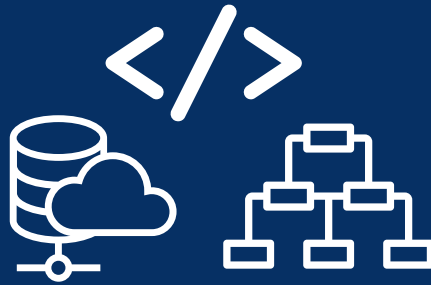
**Software Workflows**



**Integrations**



**Ease of Use**



**Software Workflows**



**Integrations**

# See all the details in the Release Notes

Filter results by:

- Release range
- Category
- Text filter
- Incompatibilities
- Highlights

The screenshot displays the MATLAB Release Notes interface. At the top, it says "MATLAB Release Notes" with a version indicator "R2026a" and a link to "expand all in page". Below this, there are tabs for "Bug Reports" and "Bug Fixes". A section titled "YOUR SELECTIONS" shows "Performance" selected. A yellow box highlights "Found 81 notes" and "Release Range: R2023a to R2026a". A dropdown menu for "Performance" is open, showing a list of categories: Environment, Language and Programming, Data Analysis, Data Import and Export, Mathematics, Graphics, App Building, Performance (checked), Software Development Tools, External Language Interfaces, and Hardware Support. A text filter box contains "MATLAB Release Notes". The main content area shows the "R2026a" release with a sub-section for "Performance" under the heading "duration Data Type: Improved performance". It lists operations on duration arrays: Arithmetic operations, Array creation, Array indexing, Concatenation, Reshaping, and Sorting. A yellow box highlights the text "about 108x faster than in the previous release." Below this, a code block shows a MATLAB function for timing tests. A black box with a white URL <https://www.mathworks.com/help/matlab/release-notes.html> is overlaid on the bottom right. At the bottom, it states "The approximate execution times are:" followed by "R2025b: 12.93 s" and "R2026a: 0.12 s".

**MATLAB Release Notes** R2026a expand all in page

Bug Reports | Bug Fixes

YOUR SELECTIONS Performance x

Found 81 notes Release Range: R2023a to R2026a Share

R2023a to R2026a Performance Incompatibilities Highlights Sort Release: Latest to Earliest

Text Filter: MATLAB Release Notes

▼ R2026a New Features, Bug Fixes, Compatibility Updates

**Performance**

▼ duration Data Type: Improved performance

Operations on duration arrays supported:

- Arithmetic operations
- Array creation
- Array indexing
- Concatenation
- Reshaping
- Sorting

For example, this code creates a duration scalar. The code is about 108x faster than in the previous release.

```
function timingTest
for i = 1:1e6
    d = duration(0,0,30);
end
end
```

<https://www.mathworks.com/help/matlab/release-notes.html>

The approximate execution times are:

R2025b: 12.93 s

R2026a: 0.12 s

# Live Editor

## Make scripts readable

- Richly formatted text
- Interleave code & output

## Add interactivity

- Hyperlinks and table of contents
- Live Controls to adjust parameters

## Share with non-MATLAB users

- Export as report (pdf, docx, etc.)
- Convert to Jupyter notebook
- Share through GitHub and MATLAB Online

SunriseSunset.mlx x +

### Estimating Sunrise and Sunset

We can estimate sunrise and sunset times if we know the latitude, longitude, and UTC offset. The [solar declination](#) ( $\delta$ ) is the angle of the sun relative to the earth's equatorial plane. On any given day of the year ( $d$ ), using the latitude ( $\phi$ ), the sun's declination ( $\delta$ ) and the solar time correction ( $SC$ ) we can calculate sunrise and sunset times.

| Solar Declination                                                                            | Sunrise                                                                   | Sunset                                                                    |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|
| $\delta = \sin^{-1} \left[ \sin(23.45) \sin \left( \frac{360}{365} (d - 81) \right) \right]$ | $12 - \frac{\cos^{-1}(-\tan \phi \tan \delta)}{15^\circ} - \frac{SC}{60}$ | $12 + \frac{\cos^{-1}(-\tan \phi \tan \delta)}{15^\circ} - \frac{SC}{60}$ |

#### Table of Contents

- [Estimating the Sunrise and Sunset Times](#)
- [Plot Yearly Results](#)

### Estimating the Sunrise and Sunset Times

Set the latitude, longitude, and UT offset.

1

lat = 42

2

lon = -71

3

UTCoff = -5

Estimate the sunrise and sunset times. We use the custom equationOfTime function to calculate the solar time [correction](#) (SC).

4

5

day = 1:365;  
timeCorr = equationOfTime(day)  
  
timeCorr = 1x365  
-3.7052 -4.1497 -4.5894 -5.0239 -5.4528 -5.8756 -6.2921 -6.7018 ...

6

plotSunriseSunset(lon, UTCoff, timeCorr, day, lat)

#### Sunrise and Sunset

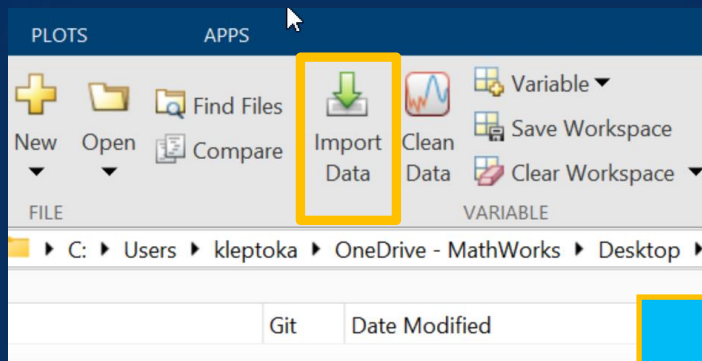
 MathWorks®

7

# Data Handling in MATLAB: Import Data Live Task

supports:

- *NetCDF (R2023b)*
- *HDF5 (R2024a)*
- *GRIB (R2024b)*
- *And more*



Code  
Generated

A screenshot of the MATLAB 'Import Data' task window. The window title is 'Import Data'. At the top, it shows the file path 'GLAH06\_634\_2103\_002\_0196\_2\_01\_0001' and a description: '= Struct of HDF5 data imported from GLAH06\_634\_2103\_002\_0196\_2\_01\_0001.H5'. Below this, there are two sections: 'Select source' and 'Select data to import'. The 'Select source' section shows the file path 'C:\Users\oghoosh\Downloads\GLAH06\_634\_2103\_002\_0196\_2\_01\_0001' and the file type 'HDF5, Size: 4.45 MB'. The 'Select data to import' section shows a tree view of the HDF5 data structure. The 'Data\_1HZ' group is expanded, and the 'DS\_UTCTime\_1' dataset is selected. To the right of the tree view, the properties of the selected dataset are displayed: '/Data\_1HZ/DS\_UTCTime\_1', Size: 1332, MaxSize: Inf, Datatype: H5T\_IEEE\_F64LE (double), ChunkSize: 38400, Filters: deflate(6), FillValue: 0.000000. Below the tree view, there are three sections: 'Specify subsetting options for datasets', 'Specify code generation type', and 'Display results'. The 'Display results' section shows the generated MATLAB code, which includes comments and code to create a structure to store the imported HDF5 data, read the data from the file, and display the results. The code is as follows:

```
% Create a structure to store imported HDF5 data
GLAH06_634_2103_002_0196_2_01_0001 = struct();

filename = "C:\Users\oghoosh\Downloads\GLAH06_634_2103_002_0196_2_01_0001.H5";

GLAH06_634_2103_002_0196_2_01_0001.Groups(1).Name = "Data_1HZ";

GLAH06_634_2103_002_0196_2_01_0001.Groups(1).Datasets(1).Name = "DS_UTCTime_1";
GLAH06_634_2103_002_0196_2_01_0001.Groups(1).Datasets(1).Value = h5read(filename, "/Data_1HZ/DS_UTCTime_1");

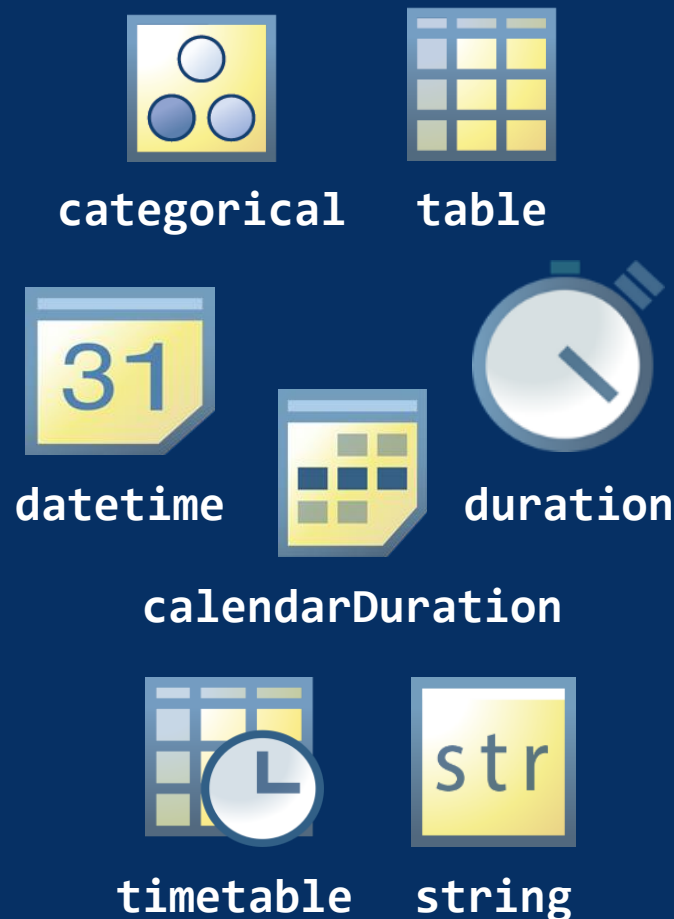
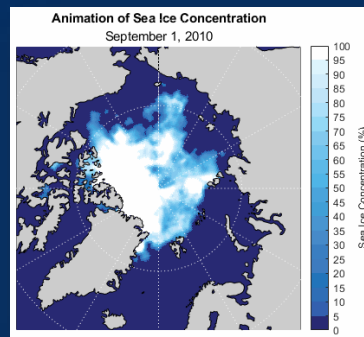
clear filename

% Display results
GLAH06_634_2103_002_0196_2_01_0001
```

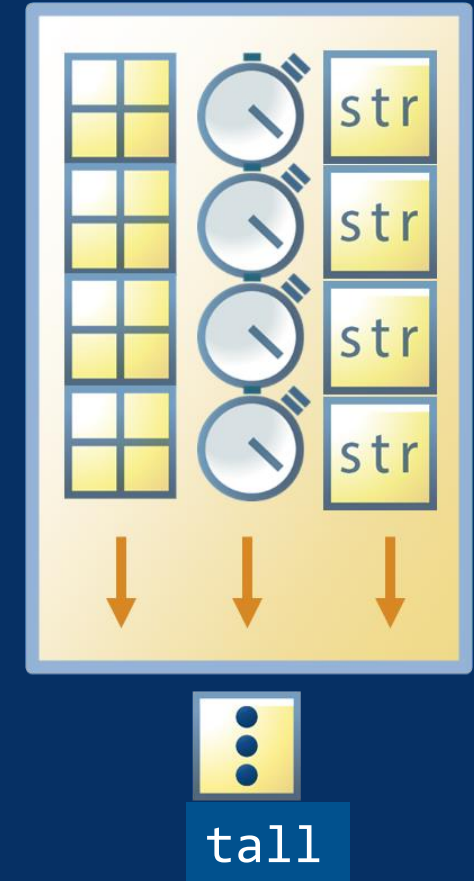
# Supported data formats and types in MATLAB

- **Raster file formats:** GeoTIFF, USGS DEM, DEM, DTED, Arc ASCII Grid, GTOPO30, ETOPO, and worldfile
- **Vector file formats:** ESRI shapefiles, KML, GPX, VMAP0, GSHHS, and .gpkg
- **Image file formats:** TIFF, JPEG, PNG, JPEG2000, LandSat, NITF and HDR
- **Scientific data:** NetCDF, HDF5, GRIB, CDF, FITS, and multiband files (BIP, BIL, BSQ)
- Community Tools to work SAC and SEG-Y

[Data Formats >>](#)



[Data Format Functions >>](#)

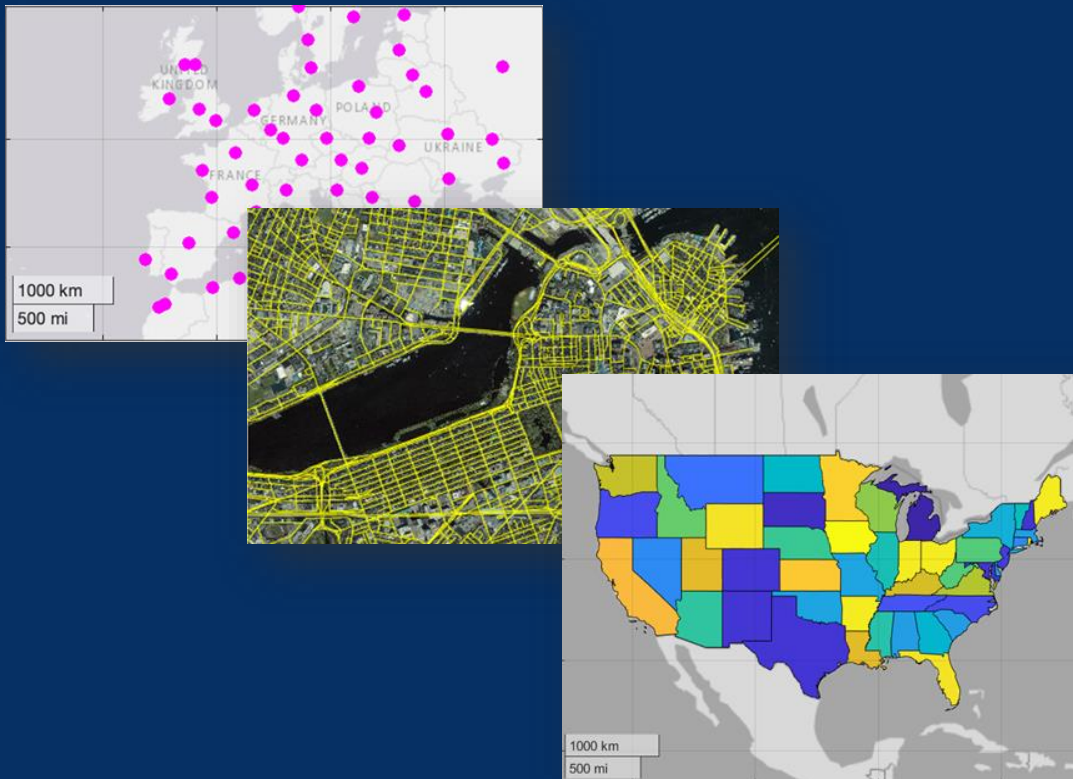


[Tall Arrays >>](#)

# Unified file readers

## Vector data

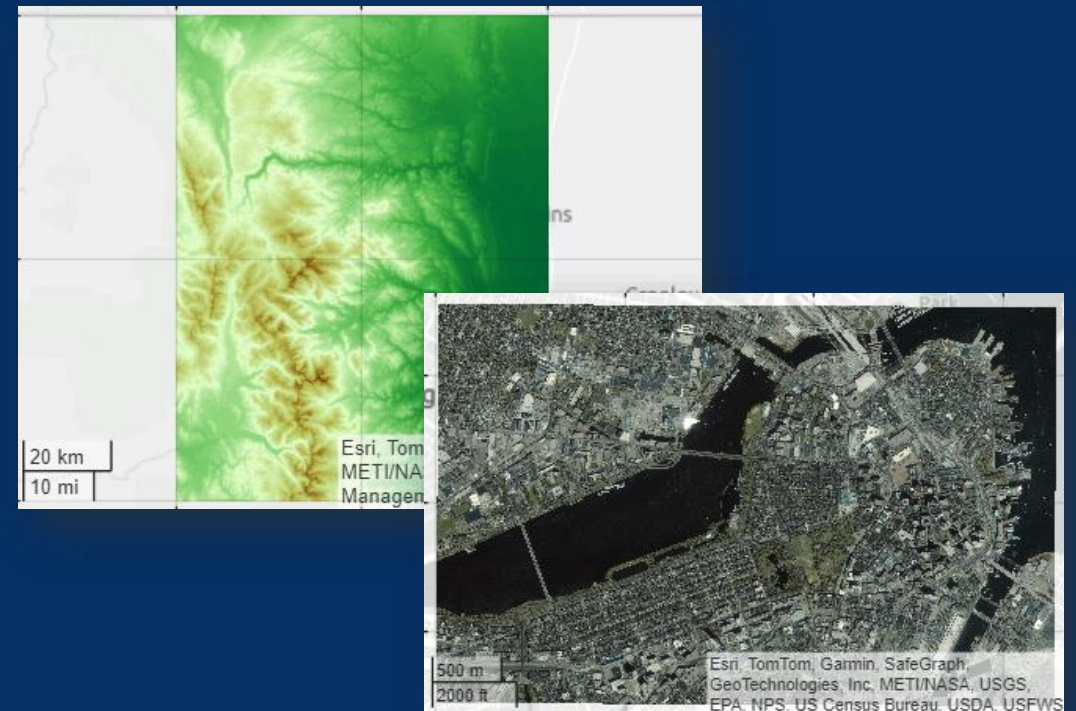
*(Geometric features - points, lines, polygons)*



```
T = readgeotable(filename);
```

## Raster data

*(Grid-based data)*

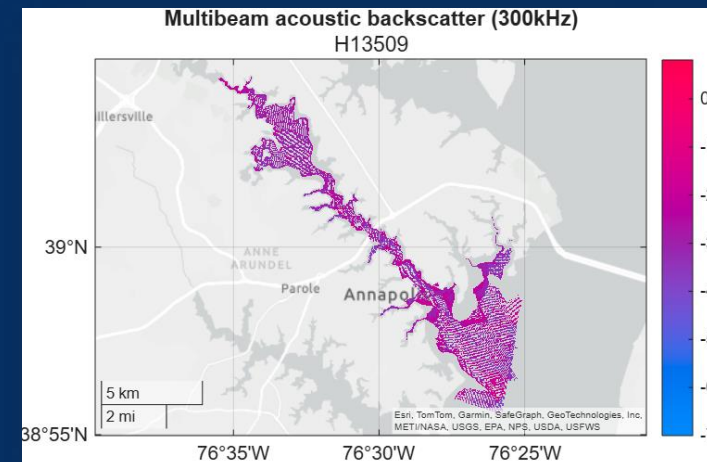
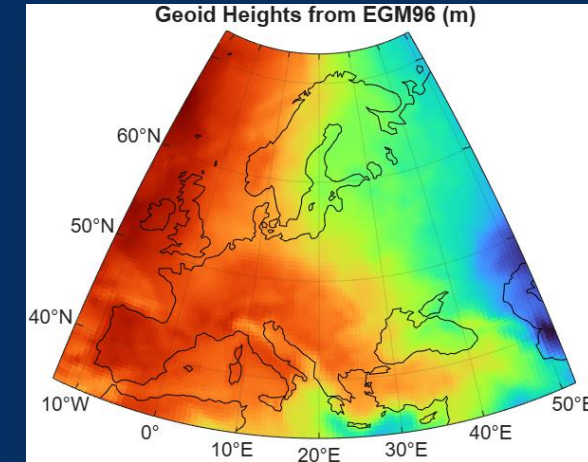


```
[A,R] = readgeoraster(filename);
```

# New and Updated Map Visualizations – 2026a

## *New Functionality*

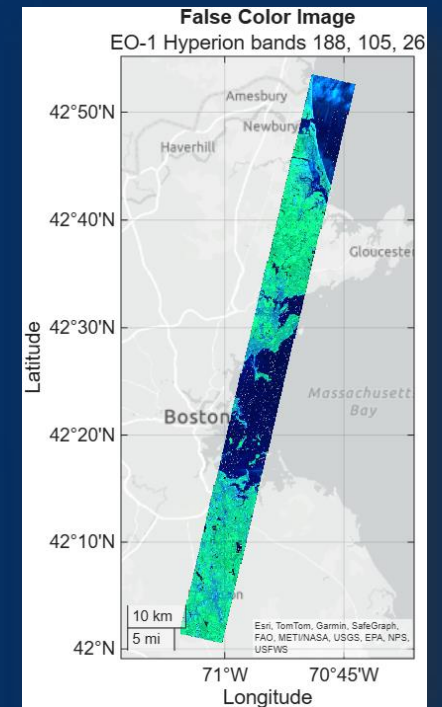
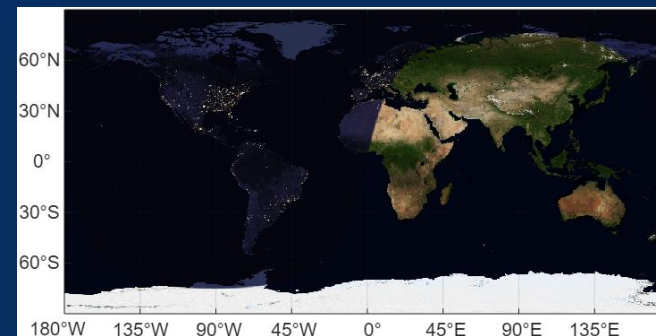
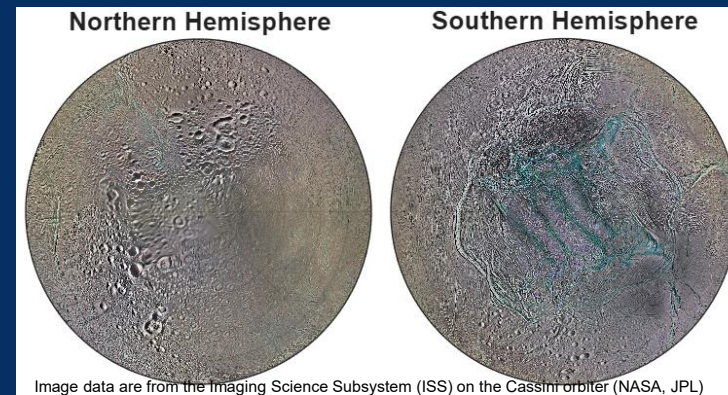
- Psuedocolor raster displays on maps



# New and Updated Map Visualizations – 2026a

## New Functionality

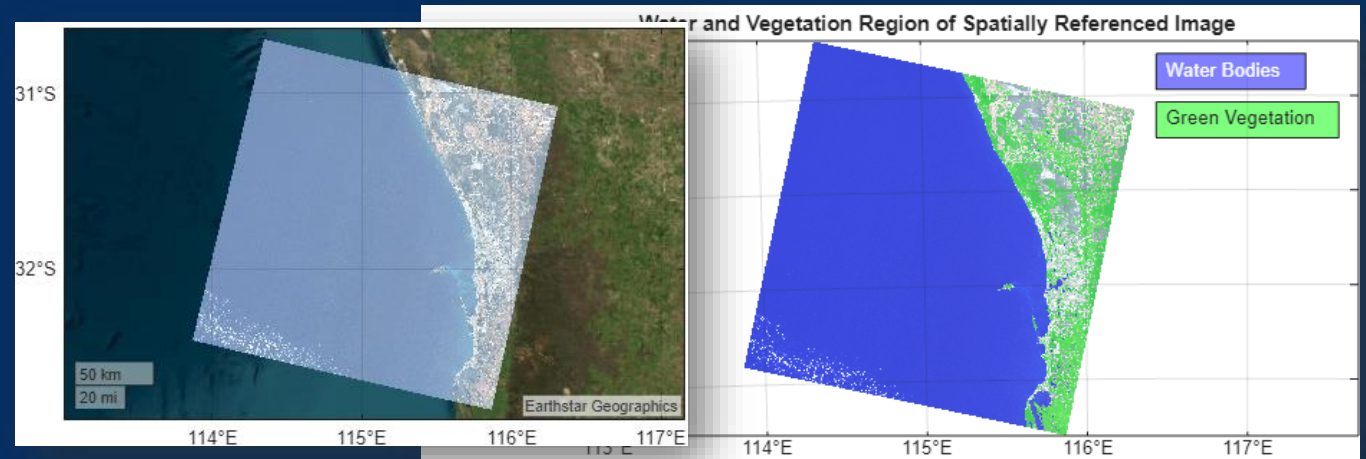
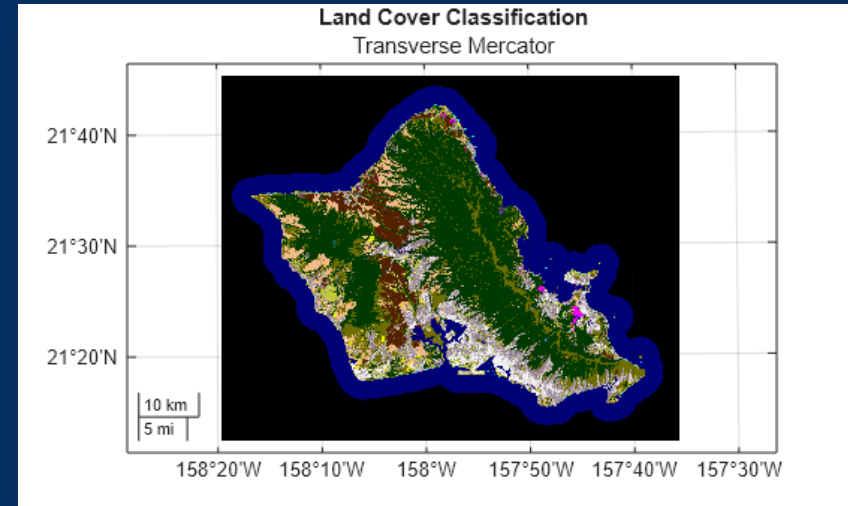
- Psuedocolor raster displays on maps
- Image displays on maps



# New and Updated Map Visualizations – 2026a

## New Functionality

- Psuedocolor raster displays on maps
- Image displays on maps
- Images and psuedocolor on map at same time
- Transparency effects on map layers

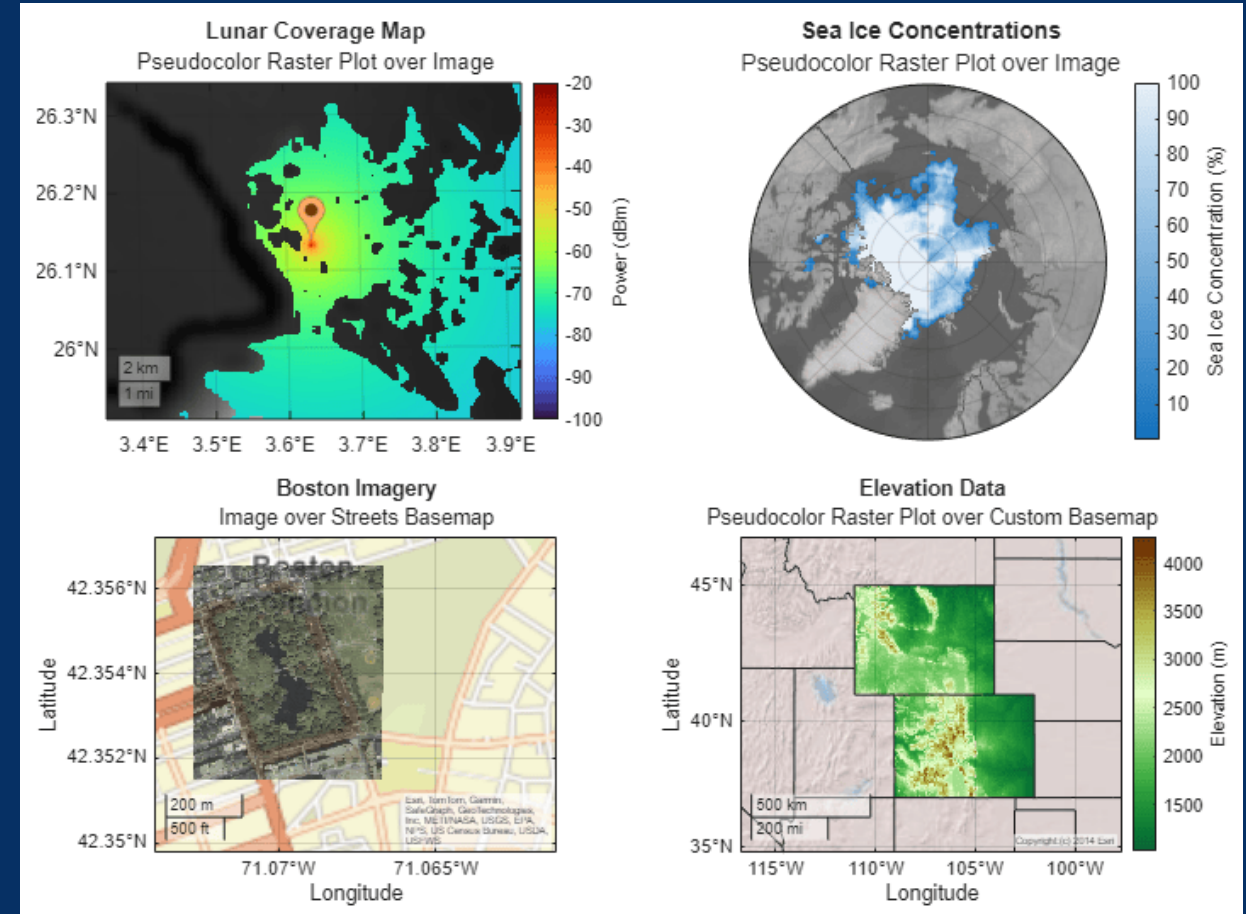


# New and Updated Map Visualizations

Demo

## Quality of Life Improvements – 2026a

- Map across international dateline/meridians
- Change projections more easily
- Redraw axes upon zoom
- Higher resolution/quality plots
- Merge multiple rasters or resample rasters
- Support for point and cell rasters

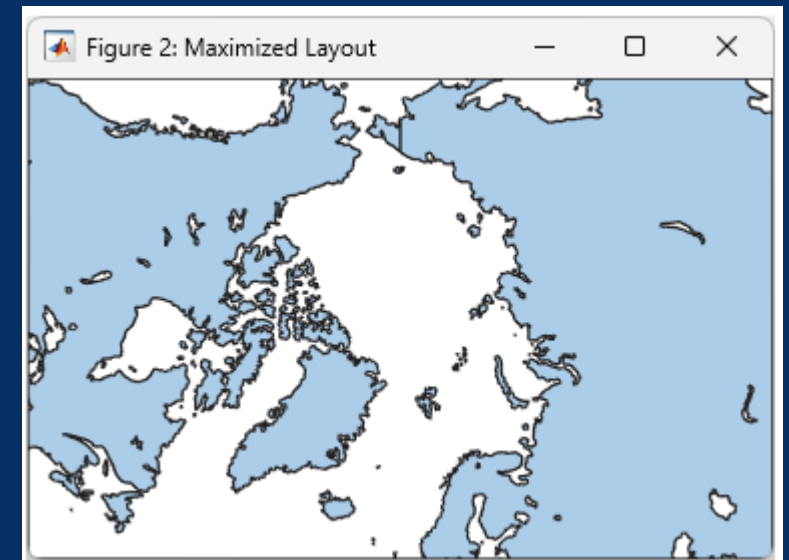
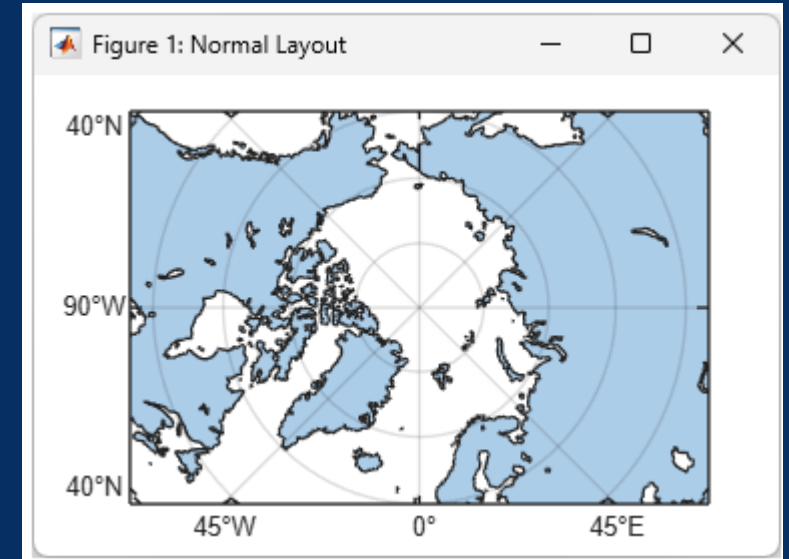


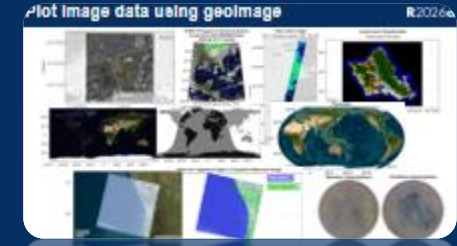
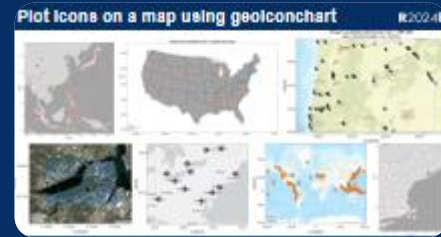
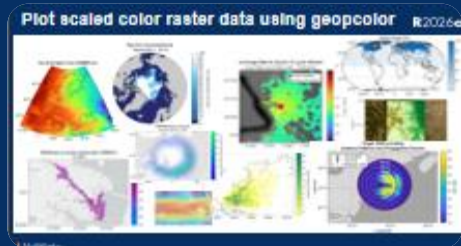
# New and Updated Map Visualizations

Demo

## *New Functionality & Quality of Life Improvements – 2026b*

- Create map axes from placename or Aol
- Maximize map axes to fill space
- 3D geographic plots: lines, markers, tooltips updated
- CRS Handling
  - Native 3D coordinate support
  - Specify CRS of raster reference objects and shape objects on creation
  - Change/update CRS of plot





# Limitless Library of Inspiring Ideas



# Where to start?

Mapping Toolbox  
Documentation & Examples

# Theme - Integrations



Ease of Use



Software Workflows



**Integrations**

# Three topics for integrations



**Languages**



**Development  
Environments**



**AI Agents**

# Three topics for integrations



**Languages**



**Development  
Environments**



**AI Agents**

# Collaborate using seven bidirectional interfaces

Co-execute MATLAB with other languages and applications

Call code and libraries written in another language from MATLAB

Bidirectional interfaces for:

- Python
- Java
- C/C++
- .NET
- REST
- COM
- Fortran



» Call MATLAB code from another language



# Example: calling python from MATLAB

- Run Python commands/scripts from MATLAB
  - py.\*** - Execute a Python command inline: `py.max(3,4)`
  - pyrun** - Execute Python statements defined as strings
  - pyrunfile** - Execute a Python script file
  - Run Python Code** Live Task - Automatically generate MATLAB code to call Python
- Interchange Python datatypes within MATLAB
  - Numeric, string, list, tuple, datetime, timedelta, dictionaries, Pandas dataframe, ...
- Control MATLAB from Python
  - See [MATLAB Engine for Python](#) documentation

The screenshot displays the MATLAB Live Editor interface. At the top, a script named 'with\_python.mlx' is open. Line 1 contains the Python code: `t = table(["a"; "b"; "c"], datetime("now")+days(0:2)')`. Below this, the output shows a 3x2 table with columns 'Var1' and 'Var2'.

|   | Var1 | Var2                 |
|---|------|----------------------|
| 1 | "a"  | 03-Sep-2024 12:35:19 |
| 2 | "b"  | 04-Sep-2024 12:35:19 |
| 3 | "c"  | 05-Sep-2024 12:35:19 |

Below the table, the 'Run Python Code' task is shown. It has a dropdown menu set to 'df' and a description: 'Run Python code using MATLAB variable t'. The 'Select input type' section has 'Code' selected. The 'Enter Python code' section contains the following code:

```
df = t
import pandas as pd
def add_column(df):
    df['new_col'] = (1,2,3)
    return df
add_column(df)
```

The 'Output options' section is expanded, showing 'Show code'. At the bottom, line 27 shows the MATLAB command `table(df)`. The output is a 3x3 table with columns 'Var1', 'Var2', and 'new\_col'.

|   | Var1 | Var2                 | new_col |
|---|------|----------------------|---------|
| 1 | "a"  | 03-Sep-2024 12:35:19 | 1       |
| 2 | "b"  | 04-Sep-2024 12:35:19 | 2       |
| 3 | "c"  | 05-Sep-2024 12:35:19 | 3       |

# Manage python environments

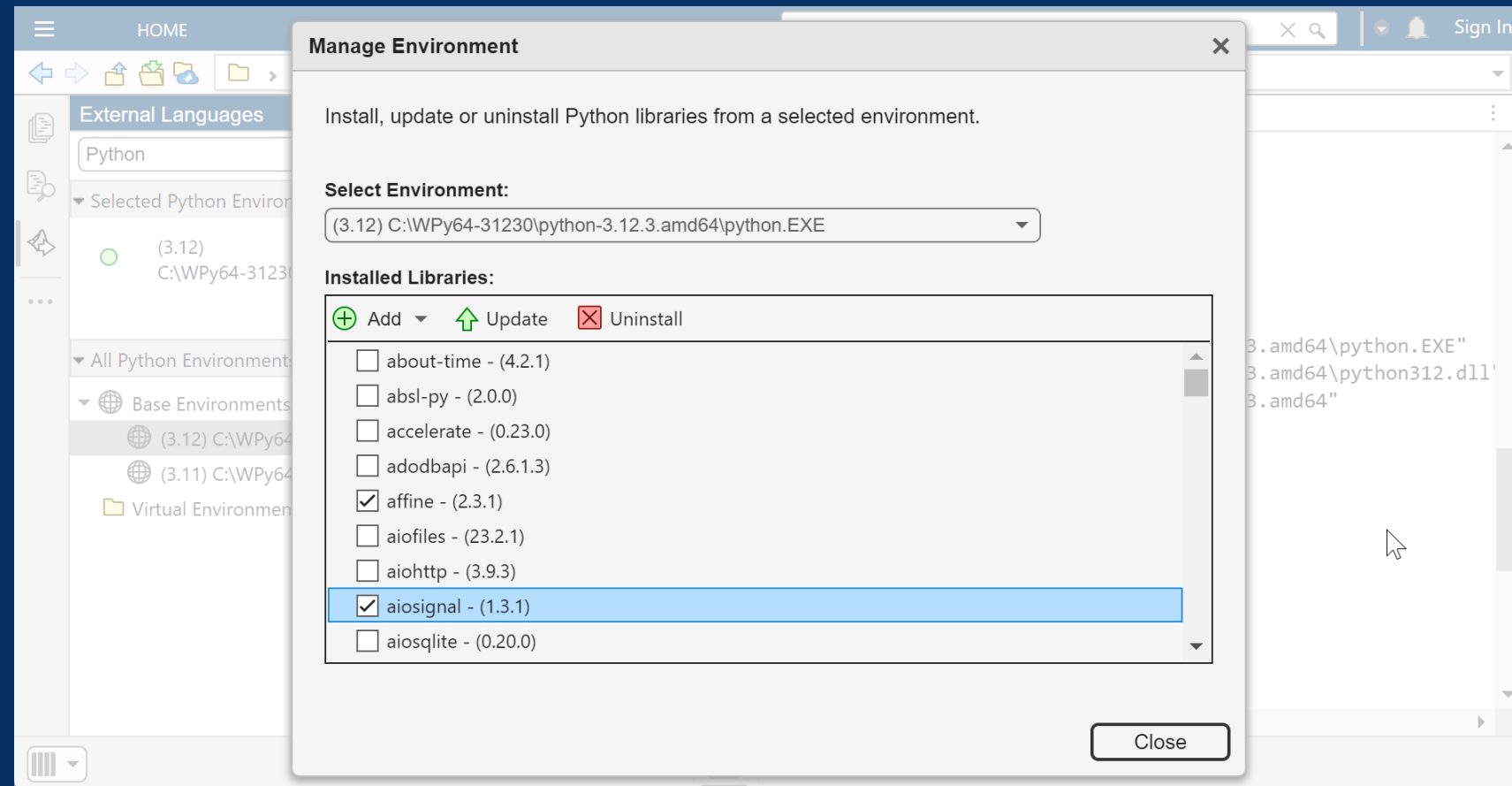
R2026a

Create and configure Python environments in the MATLAB desktop

Add or create new environments

Manage existing environments

Install, update, or uninstall libraries



# Try it out!

Demo

## Running Python with MATLAB

### Table of Contents

#### Call Python from MATLAB

##### Option 1: Setup a Python Environment

##### Converting Data Types

##### Option 2: Use the Python Live Task

##### Export MATLAB as...

##### Compile MATLAB Code as a Standalone Executable or as Python Code

#### Call MATLAB from Python

## Call Python from MATLAB

### Option 1: Setup a Python Environment

Check which version of Python is installed and that it is on your MATLAB path. If running on MATLAB online, a version of Python is already installed on the MathWorks' servers.

If multiple versions of Python are detected or installed on your computer, specify the version of Python you want to run:

If you downloaded a Python interpreter, but did not register it in the Windows registry, use:

```
pyenv(Version="[executable]") % where [executable] is the path where the int
```

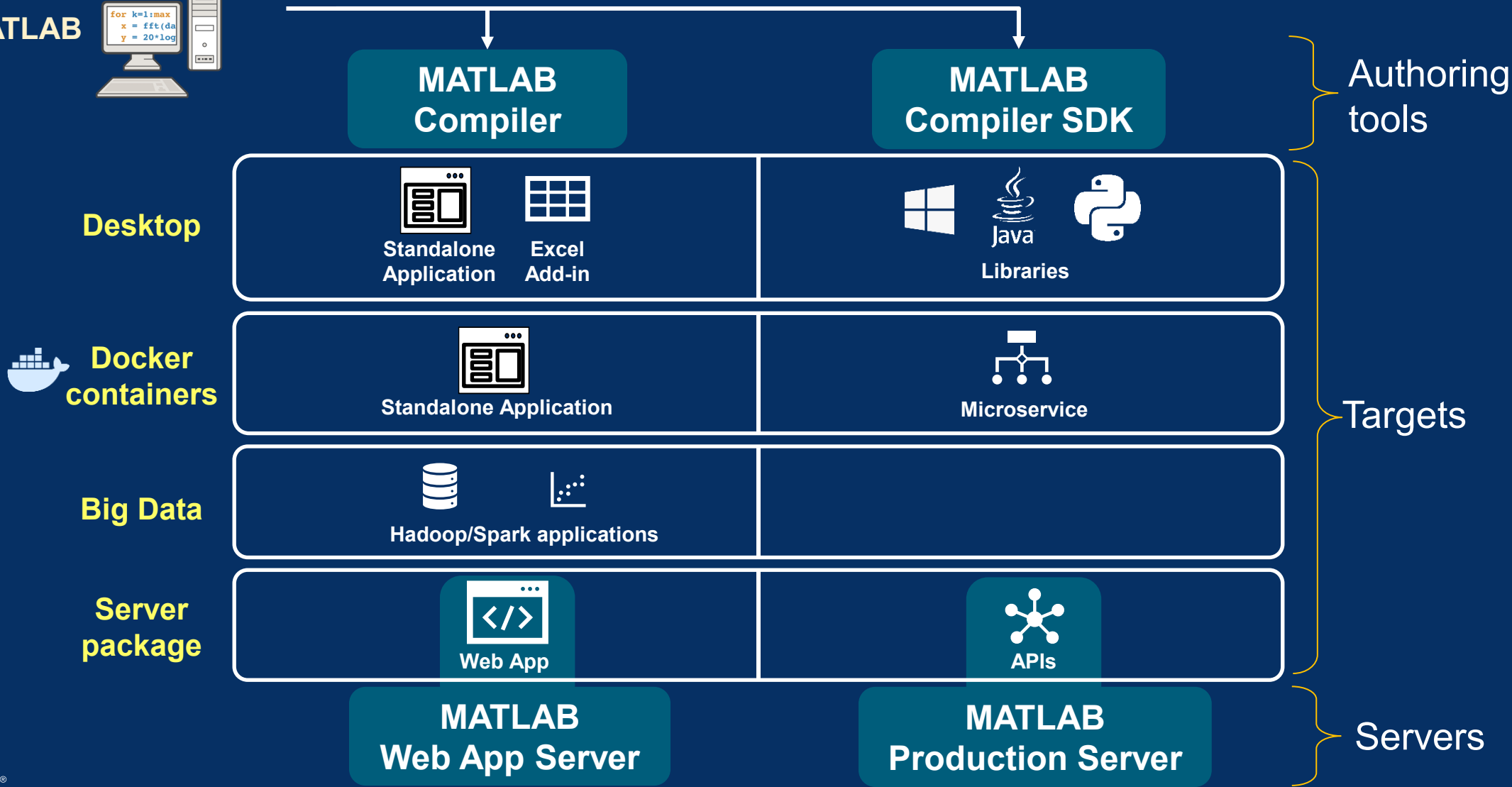
Call Python programmatically from  
MATLAB

OR

Use a Python-MATLAB Live Task

# Deploy MATLAB apps and APIs to non-MATLAB users

MATLAB

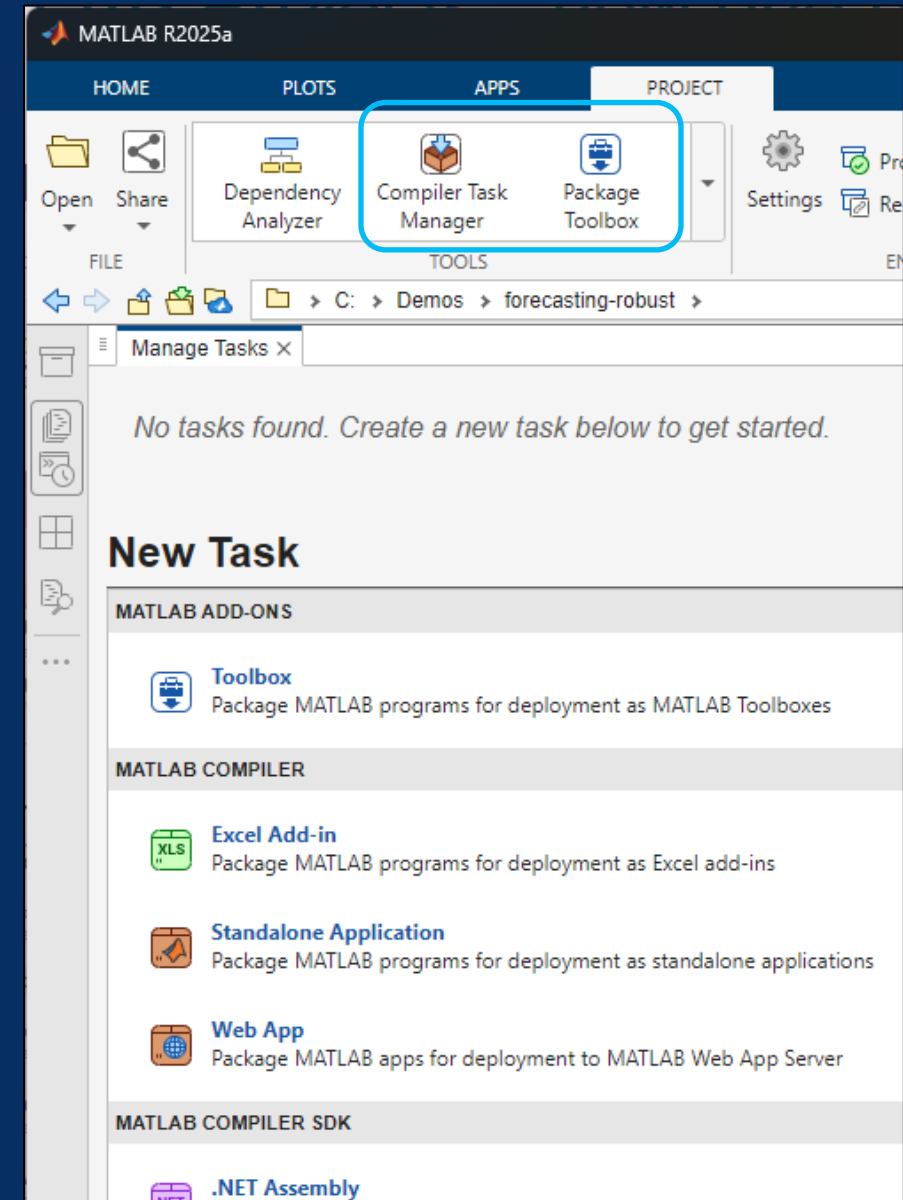


# Deployment workflows have been streamlined

App, toolbox, and MATLAB Compiler workflows have been combined

New workflow uses MATLAB Projects

Dedicated interface for each deployment target



# Integrate with AI Agents



Languages

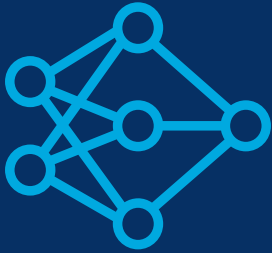


Development  
Environments



AI Agents

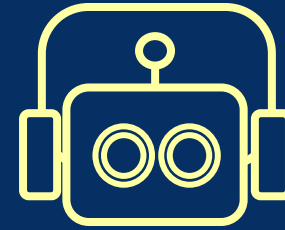
# AI in Different *Flavors*



**Discriminative AI**  
Map input to outputs to  
make predictions



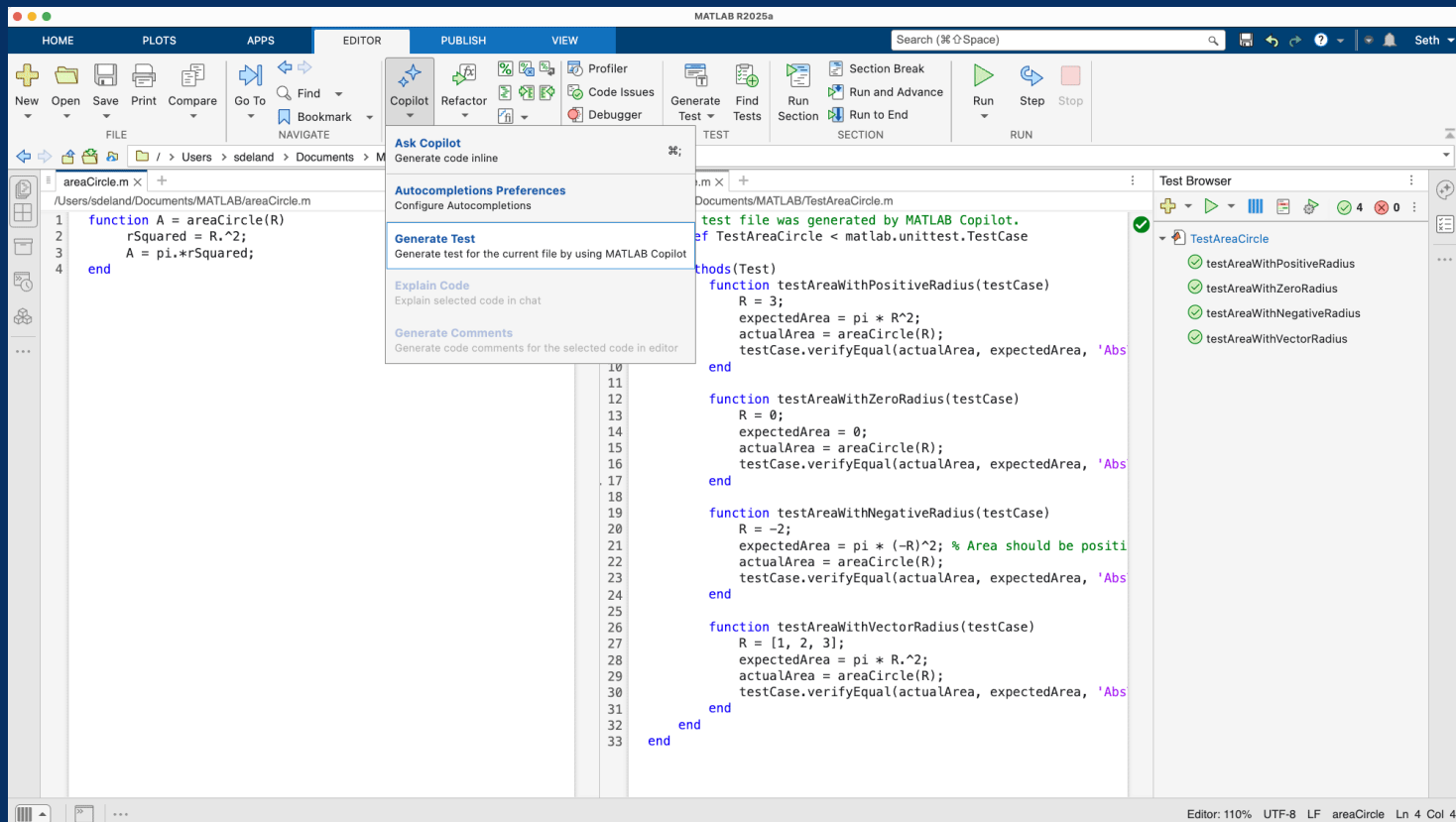
**Generative AI**  
Generate content from  
training data



**Agentic AI**  
Plan and act using tools,  
memory, and trained logic



# MATLAB Copilot



Copilot Chat



Copilot in the Editor  
& Command Window



Copilot Actions



# R2026a Simulink Copilot

## Provides deep insights and actionable advice



### Explain & Advise

- Learn tools & techniques
- Understand the model
- Search the model
- Get advice
- Troubleshoot errors

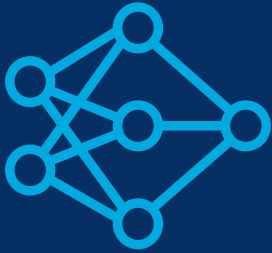


### Review & Automate

- Automate tasks
- Track progress

The screenshot shows the Simulink environment with a model titled "BatteryManagementSystem". The model diagram includes blocks for "BMS\_Algorithms", "BMS\_Info", "BMS\_Cmd", "BatteryPlantModel", and "Plant Model", connected by various signal lines. On the right, the "Copilot Chat" window is open, displaying a welcome message from Simulink Copilot. The chat window includes a text input field with the placeholder "Ask Simulink Copilot" and a "Deeper Insights" button. The status bar at the bottom indicates "Ready", "87%", and "auto(ode3)".

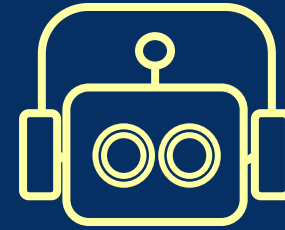
# AI in Different *Flavors*



**Discriminative AI**  
Map input to outputs to  
make predictions



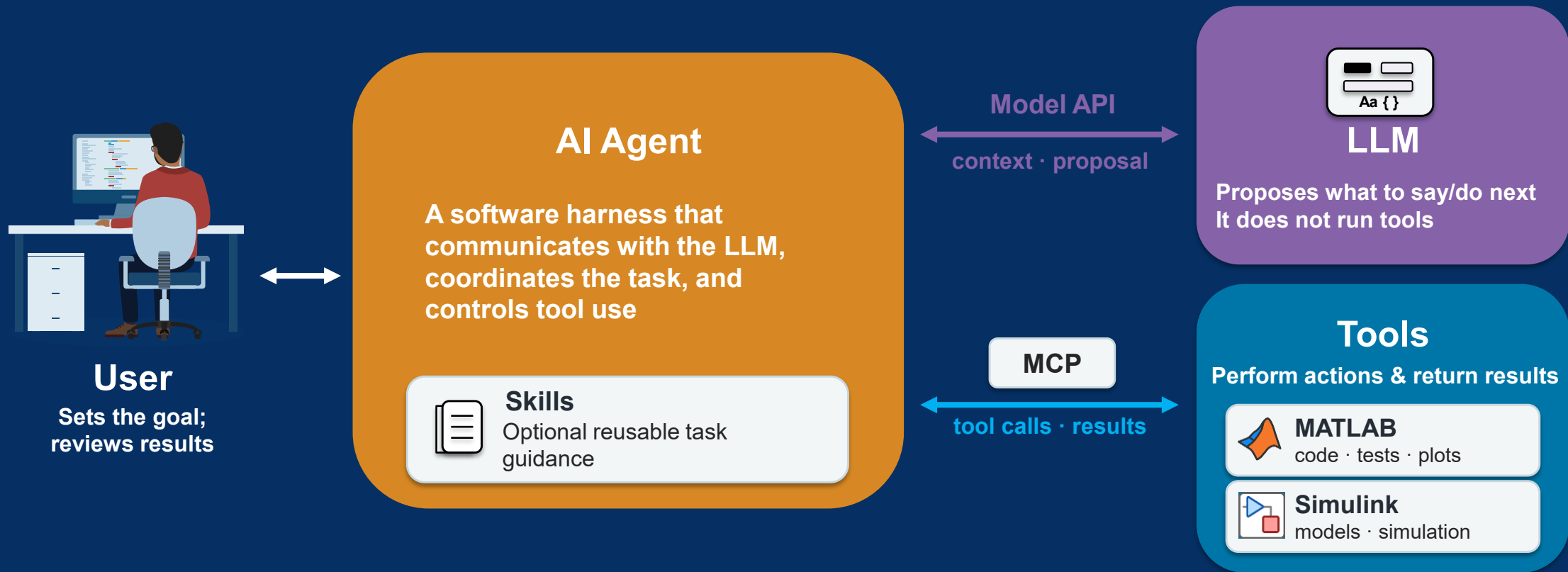
**Generative AI**  
Generate new content  
from training data



**Agentic AI**  
Plan and act using tools,  
memory, and trained logic

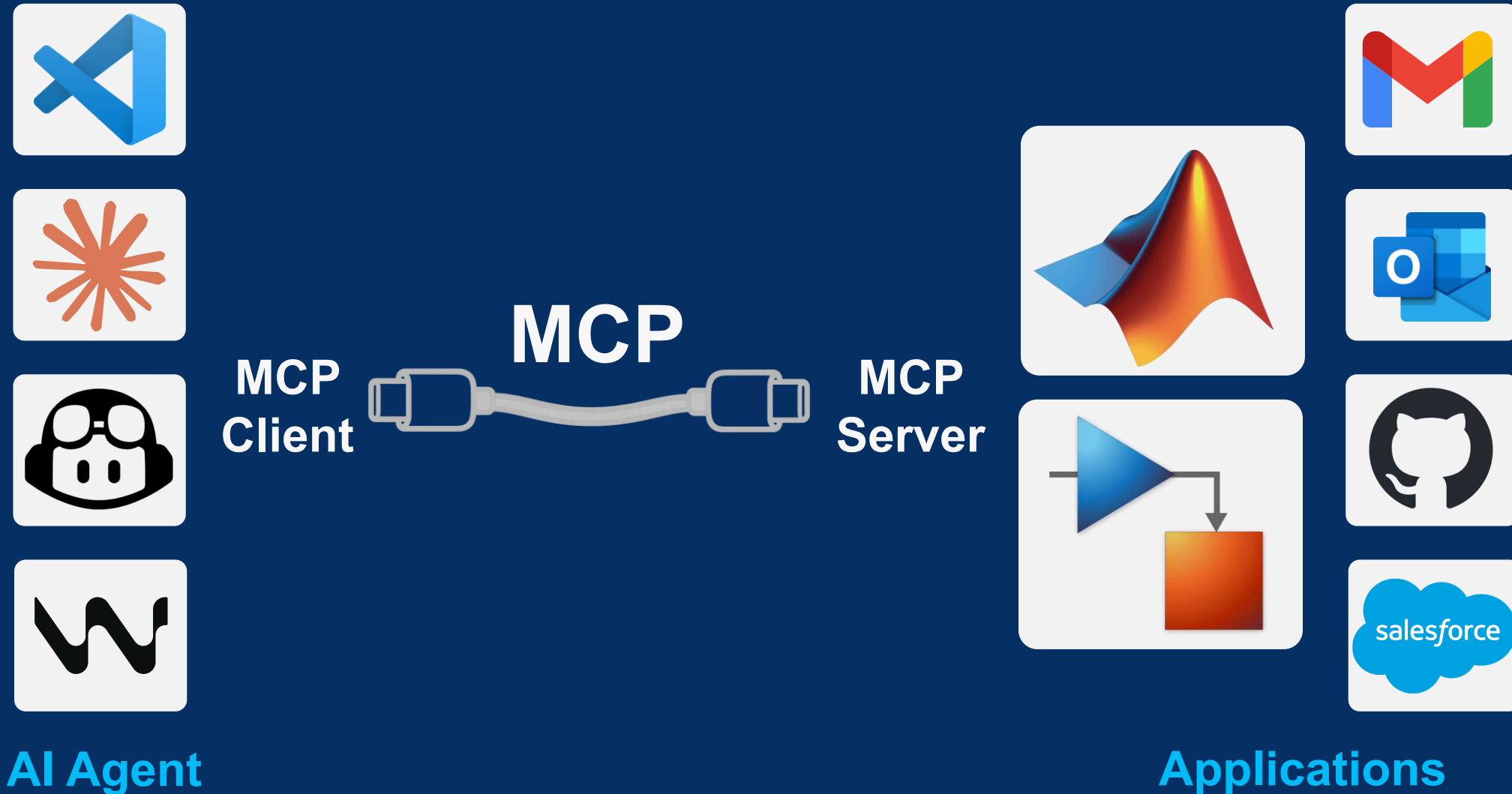


# Agentic AI: the mental model



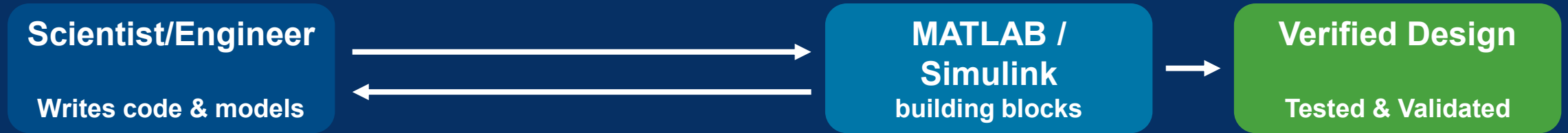
Agent coordinates · LLM proposes · Tools act · You review

# Enabled by Model Context Protocol (MCP)



# AI changes creation speed, not accountability

Today

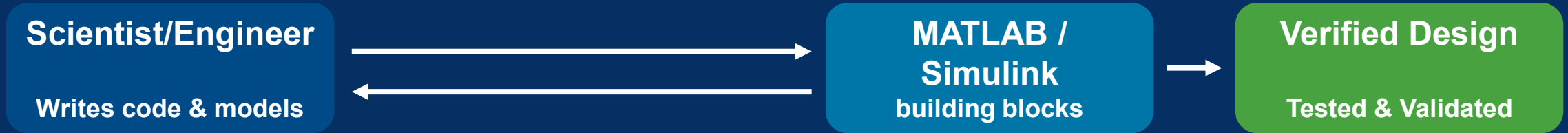


With AI



# AI changes creation speed, not accountability

Today



With AI



*What doesn't change when you add AI*

**You own the design. The same evidence verifies it.**

Speed comes from the agent.  
**Trust comes from the verified workflow.**  
Intent and creativity come from you.



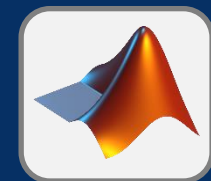
# MATLAB Agentic Toolkit

Connect trusted MATLAB/Simulink capabilities to AI agents

- A free, open-source local MCP server that connects GenAI assistants to a **locally licensed MATLAB**
- Expert Skills Catalog guides AI with best practices for testing, debugging, coding, and model creation.
- Repo + Installation instructions  
<https://github.com/matlab/matlab-agentic-toolkit>

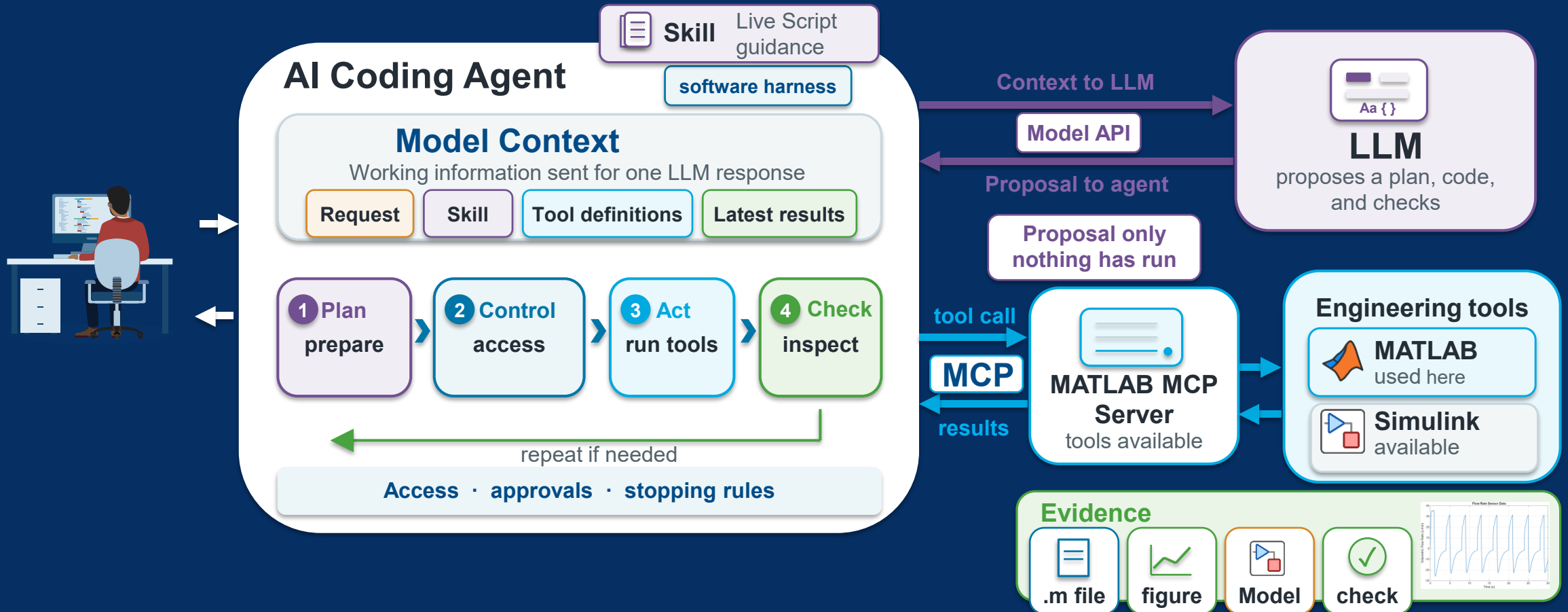


| MATLAB Skills                                                                                                                                                                                         |                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| MATLAB MCP Server                                                                                                                                                                                     |                                                                                                                                                |
| MATLAB MCP Tools                                                                                                                                                                                      |                                                                                                                                                |
| <ul style="list-style-type: none"><li>• <code>evaluate_matlab_code</code></li><li>• <code>run_matlab_file</code></li><li>• <code>detect_matlab_toolboxes</code></li></ul>                             | <ul style="list-style-type: none"><li>• <code>check_matlab_code</code></li><li>• <code>run_matlab_test_file</code></li></ul>                   |
| Resources                                                                                                                                                                                             |                                                                                                                                                |
| <ul style="list-style-type: none"><li>• <code>matlab_coding_guidelines</code></li></ul>                                                                                                               | <ul style="list-style-type: none"><li>• <code>plain_text_live_code_guidelines</code></li></ul>                                                 |
| Simulink Skills                                                                                                                                                                                       |                                                                                                                                                |
| Simulink MCP Tools                                                                                                                                                                                    |                                                                                                                                                |
| <ul style="list-style-type: none"><li>• <code>model_overview</code></li><li>• <code>model_read</code></li><li>• <code>model_query_params</code></li><li>• <code>model_resolve_params</code></li></ul> | <ul style="list-style-type: none"><li>• <code>model_edit</code></li><li>• <code>model_test</code></li><li>• <code>model_check</code></li></ul> |



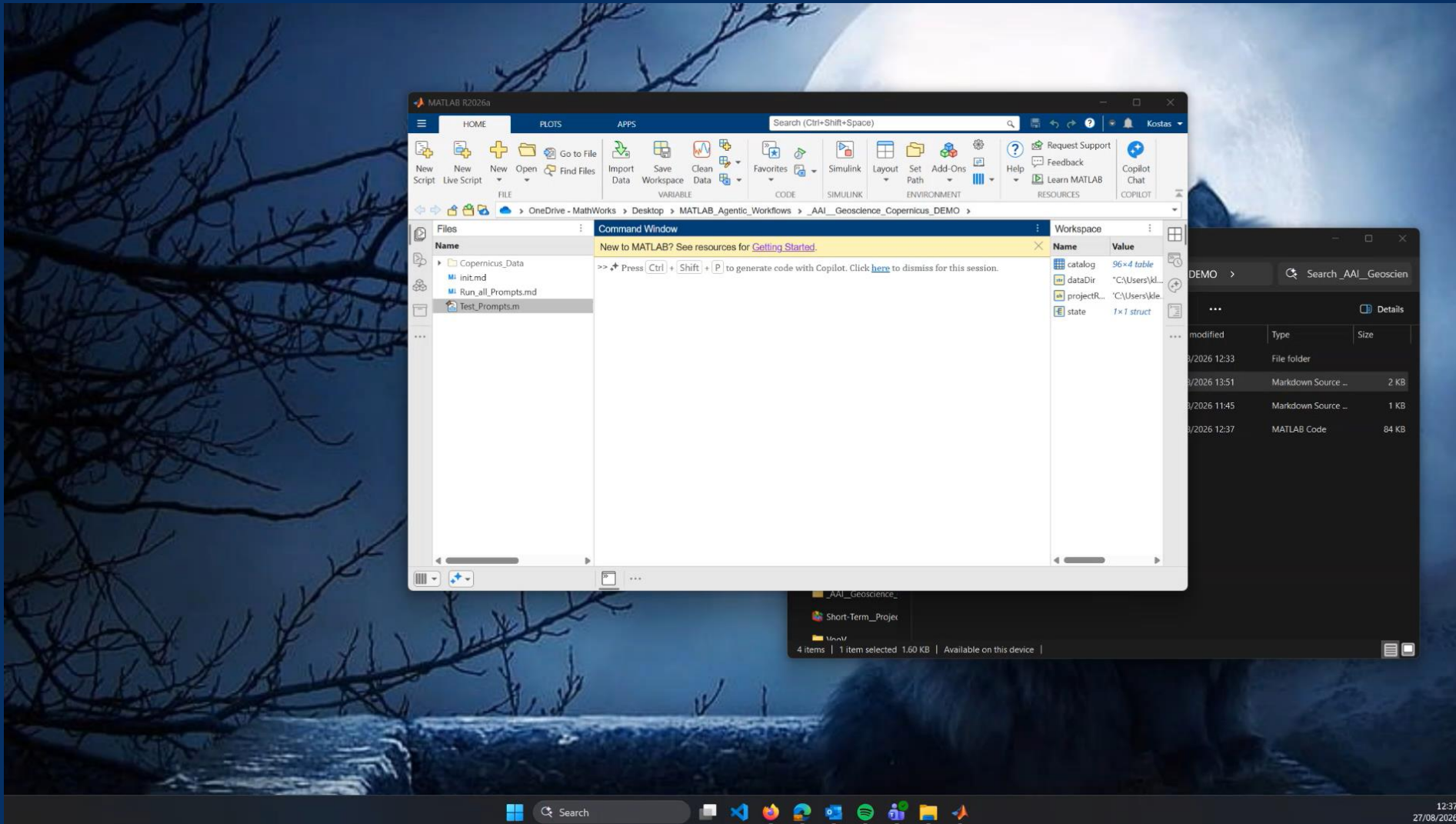
# Agentic AI in action: from request to checked result

**USER REQUEST** Create a MATLAB script that plots signal data from Simulink model



**MATLAB and Simulink turn proposals into code, simulations, plots, and tests you can inspect.**

# Agentic AI in action – Kostas's Climate Example



## Setup used

1. Agent: Codex
2. LLM: GPT 5.5
3. MATLAB R2026a
4. MATLAB Agentic Toolkit & Simulink Agentic Toolkit

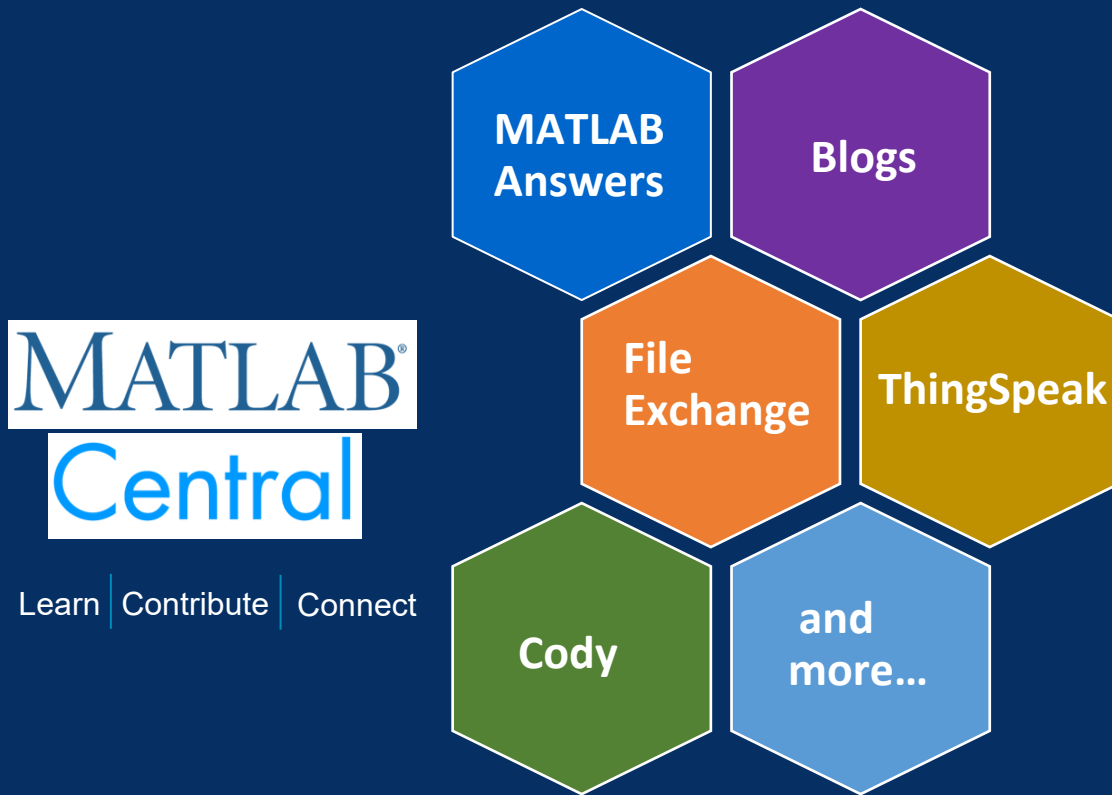
# What else can you do? **More like, what else can't you do...**

- Use other measured signals
- Modify the physical model
- Develop a new controller
- Use a different model from our documentation
- Use a wrapper to deploy app to mobile
- Use MATLAB Mobile to capture sensor data
- Add communication to local agent to interpret results

- [MATLAB Agentic Toolkit](#)
- [Simulink Agentic Toolkit](#)
- [MATLAB AI Agent SDK](#)
- [Introduction: Agentic AI with MATLAB and Simulink](#)
- [MathWorks Documentation](#)
- [File Exchange](#)

# MATLAB Central Community

Every month, over **2 million** MATLAB & Simulink users visit MATLAB Central to get questions answered, download code and improve programming skills.



**MATLAB Answers**: Q&A forum; most questions answered in < **1 hour**

**File Exchange**: Download code from a huge repository of free code including **tens of thousands** of open source community files

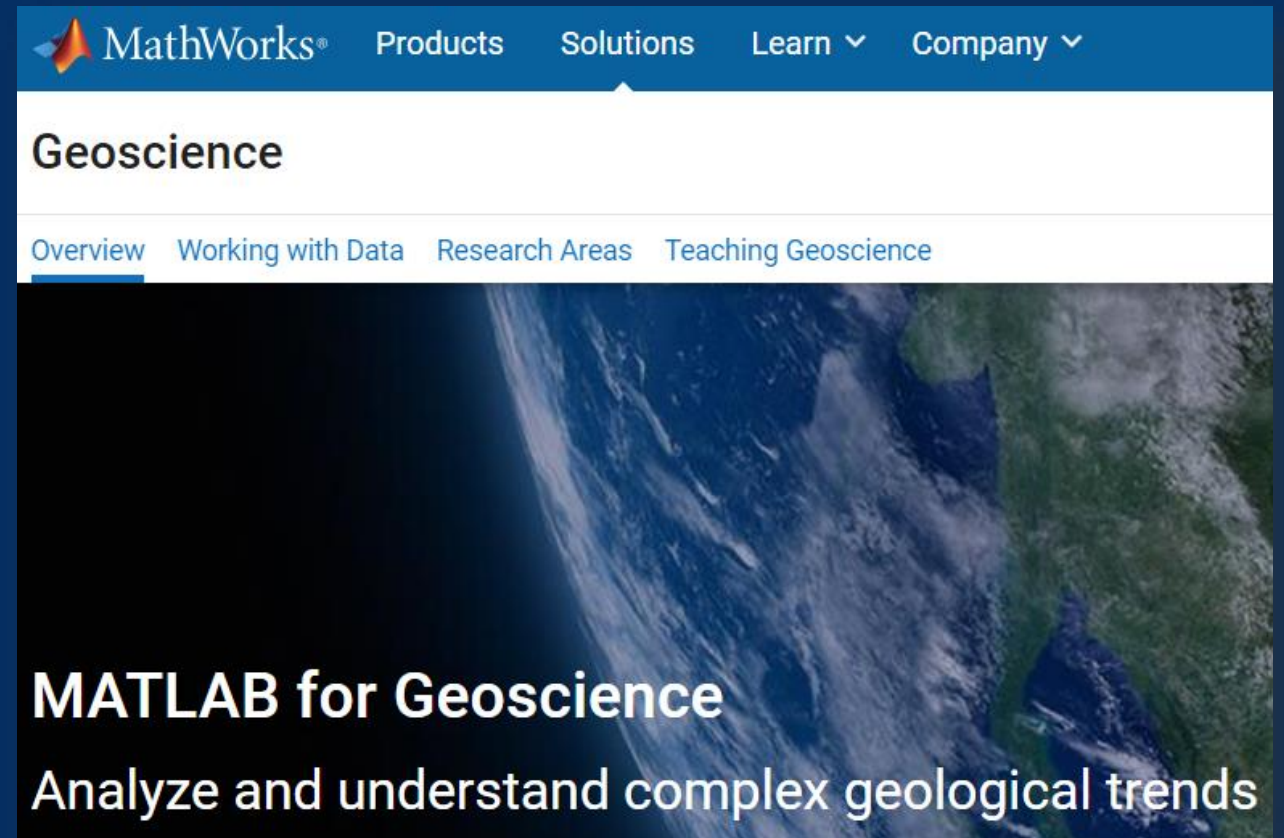
**Cody**: Sharpen programming skills while having fun

**Blogs**: Get the inside view from Engineers who build and support MATLAB & Simulink

**ThingSpeak**: Explore LoT Data

# Check out the MATLAB for Geoscience webpage

- [Geoscience Overview](#)
- [Working with Geoscience Data](#)
- [Research Areas](#)
  - Seismology and Geophysics
  - Ocean and Climate
  - Hydrology and Water Science
  - Agriculture
- [Teaching Geoscience](#)





Follow MathWorks at

